Step 7: Pricing the PPA and its risk (ticket PPA-7)

> **Numbers on this page.** The parks are the Borrentin solar park (114 MWp) and the Everswinkel wind park (28.5 MW); prices are calibrated to the market forward (EEX Cal-27 base 140.21 EUR/MWh, Cal-28 104.75, Cal-29 87.58, each quarter at the market's seasonal shape). All numbers are from `reports/step07_ppa_pricing.md` (v1.11). The updates in sections 4.6 to 4.8 explain what each later version added.
>
> **Time needed:** about 5 hours. Concepts take 75 minutes, code tour and findings 45 minutes, hands-on, notebook and exercises 2 hours, to-dos 45 minutes. **You will be able to:** price the standard PPA structures on joint scenarios, measure what each one leaves with the producer (cash flow at risk), compute the buyer's risk premium after a baseload hedge and split it into profile and volume risk, and build an indicative ten-year price from a forward price, with its sensitivities.

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0. The scenario

Tuesday, pricing committee at 3 pm. The head of origination:

> *"We have the scenarios. Now I need a number I can put in front of the buyer for ten years of the Borrentin solar park, a number I can defend in front of Risk, and one sentence on how much risk we keep. Ticket PPA-7."*

Steps 1 to 6 built the ingredients: prices that reconcile (Step 1), capture prices (Step 2), the contract cash flows (Step 3), the parks' output over 31 weather years (Step 4), price paths (Step 5) and the two simulated together (Step 6). Step 7 turns them into the answer to the project question.

Ticket PPA-7: acceptance criteria (all met today)

- ☑ Every structure (merchant, pay-as-produced with and without a negative-price clause, merchant with a floor, baseload) on the joint scenarios: fair price, expected and P5 revenue, 95% cash flow at risk.
- ☑ The buyer's risk premium after a baseload hedge, split into profile and volume risk; the unhedged premium for comparison.
- ☑ A price stack from the forward to an indicative price, with stated cost assumptions.
- ☑ A ten-year price with discounting, degradation and sensitivities (capture-rate trend, forward, analog pool).
- ☑ The decision on the analog pool from Step 6, made explicit in the configuration.
- ☑ Report, `ppa_pricing.json` for Step 9, learning page, notebook section, exercises.

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1. Concepts you need today

1.1 Two sides of the same contract

A pay-as-produced (PaP) PPA at strike K moves risk from one party to the other:

	Producer (Encavis)	**Buyer**
Receives / pays	receives K for every MWh produced	pays K for every MWh produced, sells it at P
Annual cash flow	$K \cdot \Sigma q$	$\Sigma q (P - K)$
Risk it keeps	volume only (how much the weather delivers)	price level, shape (profile) and volume

The same contract therefore needs two risk measures: the producer's **cash flow at risk** (how much less than expected could the year bring?) and the buyer's **risk premium** (how much must the strike be below the fair price for the buyer to accept the risk?).

1.2 Cash flow at risk

For annual revenue R over the scenarios,

```
CFaR95 = E[R] − q5%(R)
```

the shortfall of a bad year (one in twenty) against an average year. It is the producer's version of value at risk, in cash, which is what lenders and the CFO care about. A structure that fixes the price leaves only volume risk; a structure that fixes the band (baseload) leaves profile risk; merchant leaves everything.

1.3 The buyer's premium, and why the hedge matters

At the fair price K *the buyer breaks even on average:* $E[\Sigma q (P - K)] = 0$. Its risk is the spread around zero. The premium it asks for is the 5% shortfall per expected MWh:

```
premium = (E[V] − q5%(V)) / E[Σ q],      V = Σ_t q_t (P_t − K)
```

But most of that spread is the **price level**: if 2027 is a 160 EUR/MWh year instead of 140, the buyer gains on every MWh. The level is hedgeable with a baseload forward: the buyer sells a band $B = E[\Sigma q] / 8{,}760$ MW at the forward F, adding $B \cdot \Sigma_t (F - P_t)$ to V. What is left is the risk nobody can hedge with standard products: **profile** (the shape of the output against the shape of prices) and **volume** (how much the weather delivers, and the fact that sunny and windy years are cheap years). That residual premium is what the buyer can reasonably ask for; the unhedged number would charge the producer for a risk the buyer can remove at almost no cost.

Splitting it:

- **profile:** the same calculation with the *expected* hourly volume in every scenario (only the price shape varies);
- **volume:** joint − profile: what varying volumes, and their correlation with prices, add.

1.4 The price stack

```
indicative PaP price = forward × capture rate − risk premium − balancing − credit
                     = fair price            − risk premium − costs
```

The forward and the capture rate come from the scenarios (Steps 5 and 6); the premium from 1.3; balancing and route-to-market costs and credit and liquidity costs are stated assumptions (`[ppa.parks.*]` in the configuration: balancing 2.18 EUR/MWh for solar and 1.55 for wind, measured from imbalance prices since v1.10 plus a 0.5 marketing fee; credit 1.0 for both, an assumption).

1.5 From one year to ten

The scenarios cover 2027. A ten-year price needs assumptions for the later years:

```
K_y = F_y · c_y,   c_y = c_2027 + drift · (y − 2027) + γ(F_y) − γ(F_2027),
E_y = E_2027 · (1 − degradation)^(y − 2027)
K_10y = Σ_y D_y E_y K_y / Σ_y D_y E_y,   D_y = (1 + r)^−(y − 2027 + 0.5)
```

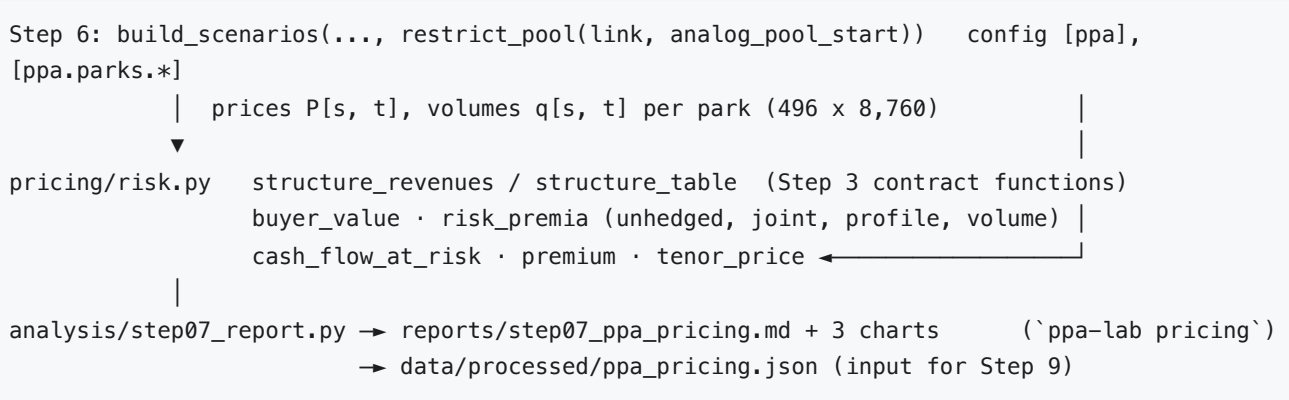
an energy- and discount-weighted average of the yearly fair prices. γ(F) is the capture rate at price level F: since v1.5 the scenarios are rebuilt at each year's forward, so a year at 88 EUR/MWh gets the capture rate of an 88 year, not 2027's (Step 8 page, section 4.8). With a flat forward and a flat capture rate it equals the 2027 price; the sensitivities show what a trend does. Since v1.11 solar's capture rate is also multiplied by a build-out factor per year (section 4.8), and its drift is 0. The risk premium and the costs per MWh of the first year are applied to every year (a simplification: a ten-year contract averages weather years but adds uncertainty about the level and the capture-rate trend).

1.6 The analog-pool decision

Until v1.7 which historical days stood in for 2027 moved the solar fair price by several EUR/MWh: recent days have deeper midday dips (Step 6, section 4.3). Step 7 takes the last twelve months as the base case (`analog_pool_start = 2025-10-01`), because 2027 will have at least as much solar capacity as 2026, and reports the whole window as a sensitivity. Since v1.7 the historical days are moved to 2027 by the measured build-out trend, and the choice no longer matters (Borrentin 2027: 77.53 against 77.47 EUR/MWh). It stays a judgement, written down where Risk can see and challenge it.

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2. What was built: architecture



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3. Code tour

3.1 `pricing/risk.py`

- `structure_revenues` reuses the **Step 3 contract functions unchanged** (`merchant`, `pay_as_produced`, `collar`, `baseload`) on (scenario × hour) arrays: the same tested code that priced history now prices the future. Each structure is struck at its fair price; the floor's fair premium is deducted from the floor variant.
- `structure_table` turns each revenue array into expected revenue, P5, CFaR95 and CFaR as a share of expected revenue.
- `buyer_value` and `risk_premia` implement 1.3; `tenor_price` implements 1.5 and returns the year-by-year table with the tenor price in `.attrs["price"]`.

3.2 `analysis/step07_report.py` and `ppa-lab pricing`

Builds the Step 6 scenarios twice (the configured pool and the whole window), prices both parks, writes the structure, premium, stack and tenor tables, the answer paragraph, three charts and `ppa_pricing.json`. About a minute: the capture rate's response to the price level rebuilds the scenarios at twelve forwards (v1.5). `ppa-lab all` now ends with this step.

3.3 Tests: what each one protects

Test file	Protects against
`test_risk.py` (14)	(v1.8) a merchant that does not curtail, a clause struck at the plain strike, a deemed strike that misses the value of curtailing, a tenor that ignores the delivered volume; wrong CFaR or premium arithmetic; a buyer that does not break even at the fair price; a hedge that does not remove the level risk of flat volumes; a profile/volume split that does not add up or misses the volume–price link; structures with different expected revenue at fair strikes; a free floor; a clause that ignores negative prices; tenor prices without drift, degradation or discounting; a forward curve that is ignored; a capture rate that does not move with the level, or is read off the wrong scale (it is linear in ln level, flat beyond the measured range)
`test_level_effect.py` (6)	a regression that misses a known slope or invents one, a sample that does not start where configured, capture rates and shares computed wrongly, an answer where the data cannot separate level from build-out, a build-out pace from incomplete years
`test_step07_report.py` (2), `test_config.py` (+1), `test_cli.py` (+1)	the report end to end and its internal consistency (indicative = fair − premium − costs), the missing-weather message, `[ppa]` validation, `ppa-lab pricing` and `all`

Planted bugs that the tests catch: the joint premium without the hedge, the profile premium with the actual volumes, the floor premium not deducted, no discounting, the clause ignored. One test needed a fix to catch its bug: the first synthetic prices were never negative, so a free floor looked fair; the structures test now shifts prices into negative territory.

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4. Five real findings

4.1 A PPA removes 54-77% of the producer's risk

2027	Merchant CFaR95	PaP CFaR95	Reduction
Borrentin	EUR 3.02 m (31% of revenue)	EUR 0.70 m (7%)	77%
Everswinkel	EUR 2.10 m (25%)	EUR 0.96 m (12%)	54%

What is left is volume: a dull year still delivers fewer MWh, and for wind that is most of the risk. A baseload PPA sits in between (Borrentin EUR 1.60 m): it fixes the level but leaves the producer with the profile. Stopping at negative prices, a free floor at 0 EUR/MWh, lowers the merchant's CFaR only from EUR 3.24 m to 3.02 m: it removes the negative-price tail, not the price level.

4.2 Most of the buyer's risk is hedgeable

EUR/MWh	Unhedged premium	After baseload hedge	of which profile	of which volume
Borrentin	25.44	**10.54**	10.11	0.43
Everswinkel	34.81	**5.49**	1.95	3.55

59-84% of the buyer's risk is the price level, which a baseload forward removes. Without the hedge the "premium" would eat a third of Borrentin's fair price (25.44 of 77.53). The residual tells the story of each technology: **solar's risk is profile** (how deep the midday dip is in a given year), **wind's risk is volume** (how much wind, and the fact that windy years are cheap).

4.3 The negative-price clause costs solar a sixth of its output

In 2027 17.5% of Borrentin's output falls in hours with negative prices (Everswinkel 4.4%), and about 30% over ten years, because later forwards are lower. At the plain fair strike a clause that pays nothing for that output would cut Borrentin's revenue by the same 17.5%. History agrees: 23.6% of its modelled output fell at negative prices in 2025. Stopping at negative prices, a floor at 0 EUR/MWh, is worth 3.10 EUR per MWh of output for Borrentin (Everswinkel 0.69). A buyer that asks for the clause must pay a higher strike for it: 97.76 instead of 77.53 EUR/MWh in 2027 (section 4.6).

4.4 The indicative price, and what moves it

Ten-year price 2027-2036 (EUR/MWh)	Borrentin fair	Borrentin indicative	Everswinkel fair	Everswinkel indicative
Base case	**35.09**	**21.37**	**86.64**	**78.60**
Capture-rate trend 2 pp/yr steeper	28.07	14.35	79.55	71.51
Capture-rate trend 1 pp/yr steeper	31.58	17.86	83.09	75.05
Capture-rate trend 1 pp/yr flatter	38.59	24.87	90.18	82.14
Forward −10%	29.52	15.80	77.22	69.18
Forward +10%	40.59	26.87	96.02	87.98
Analog days from the whole window	35.06	21.34	87.06	79.02
Build-out: recent pace (16.5 GW solar, 6.5 GWh storage a year)	33.73	20.01	–	–
Build-out: EEG solar target pace (24 GW, 12 GWh)	31.64	17.92	–	–
Build-out: storage boom (17 GW, 35 GWh)	43.80	30.08	–	–
Build-out: slowdown (12 GW, 8 GWh)	36.66	22.94	–	–

The base case follows the forward curve (141 in 2027, 88 from 2029), moves each year's capture rate with that year's forward (section 7 of the report) and, for Borrentin, follows the central build-out scenario: 17 GW of solar and 12 GWh of storage a year (section 4.8). For Borrentin the biggest levers are the **solar and storage build-out** (−3.4 to +8.7), the **forward** (−5.6 / +5.5 for ±10%) and a **capture-rate trend** on top (±3.5 per pp a year). The risk premium (10.5) and the costs (3.2) together take 13.7 off the fair price. Which historical days represent 2027 no longer matters (−0.03). Everswinkel depends on the forward (−9.4 / +9.4) and on wind's own cannibalisation (±3.5 per pp a year).

4.5 Small ones

- **A deprecated plotting argument** (`vert=False` in Matplotlib 3.11) raised a warning in the report test; it now uses `orientation="horizontal"`.

- **Chart labels hit the titles** in the first price stack; the y axis now leaves room.
- **A test set-up that could not fail:** the first synthetic prices had the same daily shape in every scenario, so profile risk was zero by construction. Each scenario now has its own dip depth.
- **The analog pool** is now a configuration value with a comment that explains it, not a hidden choice in code.

4.6 Update (v1.8): who curtails at negative prices, and who keeps the value

Until v1.7 every park produced in every hour. A real park that sells at spot stops when the price is negative: its marginal cost is about zero and no subsidy is paid then (§51 EEG), so each MWh produced at −17 EUR/MWh costs it 17 EUR. Curtailing is a free floor at 0 on every hour, which is why the "merchant + floor at 0" row has gone: a park that can curtail would pay a premium for something it already owns.

Who curtails depends on the contract:

Contract	Who curtails	Strike (2027, solar)	Producer keeps
Plain pay-as-produced	nobody: the park is paid for every MWh	77.5	volume risk only
Negative-price clause	the park, unpaid	97.8 per delivered MWh	the value of curtailing, plus volume risk from negative hours
Deemed output	the buyer, who pays the output as if produced	80.6 = 77.5 + 3.1	the value of curtailing, volume risk only

In 2027 17.5% of the solar output falls in negative hours at an average of −17.7 EUR/MWh: curtailing saves 3.1 EUR per MWh of output. Over the tenor it is about 30% of the output, because the forward falls to 88 EUR/MWh and more hours go negative. That is why the clause's ten-year price per MWh of output, 27.3, beats the plain PPA's 21.4 by more than 3.1.

Three lessons:

1. **A contract decides who owns an option.** The physical option to stop producing is worth 3.1 EUR/MWh in 2027. A plain pay-as-produced PPA destroys it: the park has no reason to stop, the buyer no right to make it. Both the clause and deemed output keep it; they differ in who carries the volume risk.
2. **Per MWh is not per MWh.** The clause's 97.8 is per delivered MWh, the plain 77.5 per MWh of output. Compare structures per MWh of output, or in EUR per year, never by the strike.
3. **The clause moves risk back to the producer.** Sunny, cheap years have the most negative hours, so delivered volume falls exactly when output is high: the producer's cash flow at risk rises from EUR 0.70 m to 1.29 m, and the buyer's premium per delivered MWh from 10.5 to 14.6 EUR/MWh. The deemed-output design keeps the value without that risk, which is why buyers who can curtail (aggregators, battery owners) may prefer it.

History backs the numbers: the solar park's output at negative prices was 1.6% in 2022, 18.1% in 2024 and 23.6% in 2025; curtailing it would have saved 4.9 EUR/MWh in 2026 (January to September).

4.7 Update (v1.9): Everswinkel already has a hedge, from the state

Everswinkel is paid under the EEG market premium model. It sells at spot, and the grid operator adds per MWh

```
premium = max(AW − MW_month, 0)
```

AW is the park's award value from its tender, MW the month's market value of all German onshore wind. That is a **put on the market value, struck at AW, renewed every month**: when wind earns less than AW, the premium makes up the difference. A PPA would replace it, because a PPA buyer wants the guarantees of origin, which the EEG does not issue.

What the put is worth depends on how far the market value sits above the strike (AW 77 EUR/MWh assumed; the award is not published):

Year	Forward	Merchant revenue per MWh	Premium per MWh
2027	141	130.6	0.49
2028	105	95.9	4.11
2029-2036	88	79.6	8.88

Over ten years the EEG pays 93.8 EUR/MWh after the direct marketer's fee (1.6, the measured balancing cost since v1.10); a PPA at the indicative price pays 78.6. Three lessons:

1. **An out-of-the-money option is not worthless over time.** In 2027 the put is far out of the money; at the Cal-29 level it is near the money and worth 8.9 EUR/MWh. Value an option along the whole forward curve, not at today's price.
2. **Twelve options are worth more than one.** With an annual market value the premium is 5.14 instead of 7.23 EUR/MWh: monthly puts pay in a cheap month even when the year as a whole is not cheap.
3. **Compare like with like.** At the Cal-29 level the EEG's bad year (5% quantile 73.4 EUR/MWh) is better than the PPA's (69.4), yet its cash flow at risk is higher (0.99 against 0.58 m), because the EEG keeps the upside. The PPA's lower CFaR is not a better deal; it is a narrower one.

The assumption that matters most is the award value: at AW 60 the premium is worth 2.3, at 95 it is 16.3 EUR/MWh. In real work the first question would be to the asset manager: what is the park's AW, and was it a 2022 or 2023 tender?

4.8 Update (v1.11): storage decides solar's later years

Until v1.10 solar's capture rate fell by an assumed 1 percentage point a year after 2027. That number was a view, not a measurement, and it was the biggest lever after the forward. v1.11 replaces it with scenarios you can argue about: how much solar and how much storage Germany adds each year.

The chain has three links, each from data or a stated assumption:

```
share growth per year   Δs = κ · PV_GW − λ · (storage_GWh − 6.5)
κ  = (share 2025 − share 2022) / GW added 2023–2025  = 0.077 points per GW
λ  = cycles · solar fraction / load = 250 · 0.8 / 466 TWh = 0.043 points per GWh
capture-rate factor     φ_y = exp(γ · A · Δs · (y − 2027)),  γ = −5.3, A = 1.38
```

γ comes from the same monthly regression as section 7, with ln(capture rate) on the left: each point of solar's share of load lowers the capture rate by 5.3% (se 0.8). The log form fits the data better and has a sensible tail: a proportional fall never reaches zero, which a straight line does within a decade. A is the seasonality: new solar adds most share in sunny months, which carry most of solar's output.

Scenario (per year)	Solar GW	Storage GWh	Capture rate 2036 vs 2027	Solar 10-year indicative
Central (base case)	17	12	×0.50	**21.4**
Recent pace	16.5	6.5	×0.43	20.0
EEG solar target pace	24	12	×0.35	17.9
Storage boom	17	35	×0.95	30.1
Slowdown	12	8	×0.57	22.9

Three lessons:

1. **Storage is the swing factor, not solar.** Between "slowdown" and "EEG target pace" solar doubles and the price moves 5 EUR/MWh; nearly tripling storage from the central case moves it 9. Ask Risk for its storage view before it argues about solar.
2. **A view becomes a scenario when its inputs can be checked.** "−1 pp a year" cannot be checked; "17 GW of solar and 12 GWh of storage a year" can, against the Bundesnetzagentur's monthly register and the battery pipeline.
3. **The model now prices solar lower than −1 pp did** (21.4 instead of 27.4). The link was measured at solar shares up to 15% with little storage; at 25-30% flexible demand, exports and curtailment rules will matter, and they are not in it. Treat the central case as the cautious end of a negotiation, not as a market quote.

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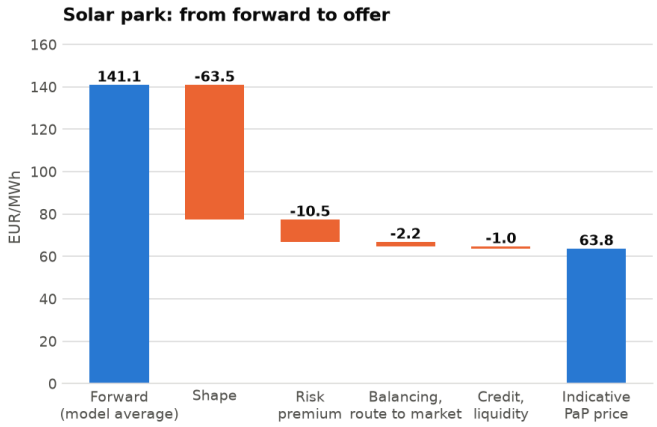
5. Hands-on: run it yourself

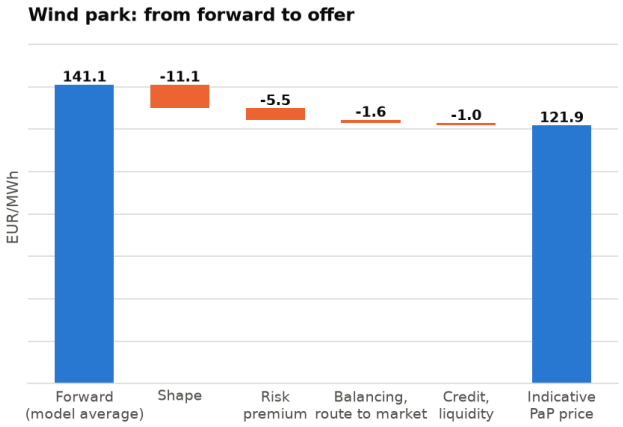
```
cd ".../Encavis quantitative/ppa-lab"
make test                  # the offline tests (290 in v1.11)
make pricing               # structures, premia, stack, ten-year price (about 12 s)
open reports/step07_ppa_pricing.md
make notebook              # Step 7 section: 7.1 to 7.6, then your solutions
```

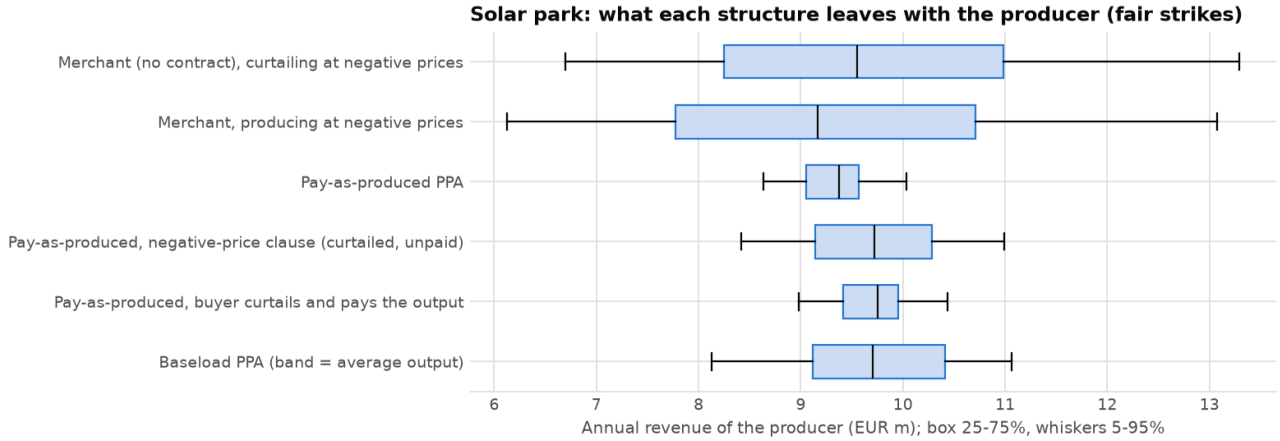
In the notebook: 7.3 shows the buyer's side at any strike, 7.5 builds your own offer, 7.6 changes the analog pool.

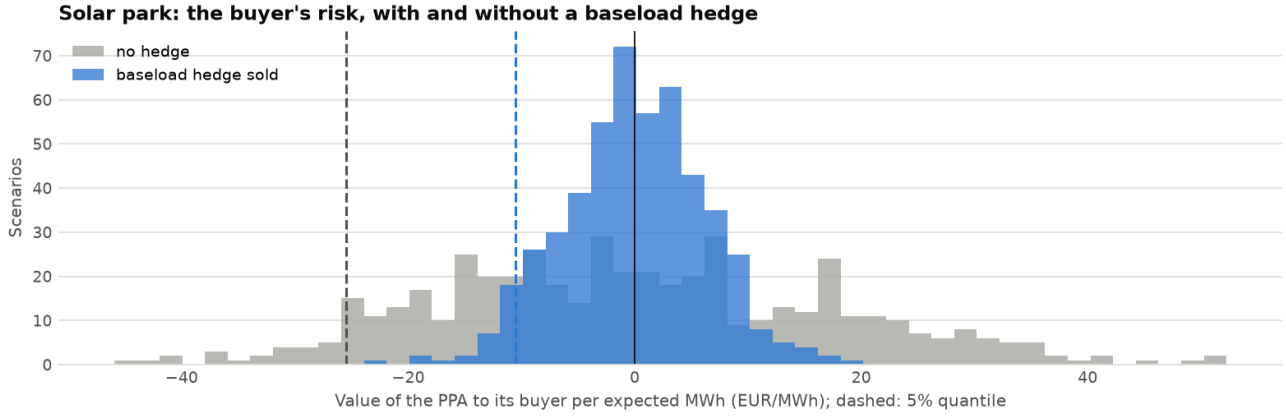
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6. Results: the answer









> *"For a 10-year pay-as-produced PPA (2027-2036) from the 114 MWp Borrentin solar park, the fair price before risk premia is 35.1 EUR/MWh (first year: forward 141.1 × expected capture rate 0.55 = 77.5; later years follow the falling forward curve and solar's build-out). After a profile and volume risk premium of 10.5 EUR/MWh and balancing and credit costs of 3.2 EUR/MWh, the indicative price is 21.4 EUR/MWh. Expected annual energy is 120.9 GWh (P50; one-year P90 113.4 GWh). Under the PPA the producer's 95% cash flow at risk falls from EUR 3.02 m (merchant) to EUR 0.70 m per year: only volume risk is left. After 2027 the capture rate follows the central build-out scenario; across the scenarios the indicative price ranges from 17.9 to 30.1 EUR/MWh, and storage decides where."*

6.1 Five stories

1. **Solar sells at about half the baseload price.** 141.1 × 0.55 = 77.5 in 2027: the shape alone takes 63.5 EUR/MWh, six times the risk premium.
2. **The hedge is the negotiation.** A buyer who quotes a 25 EUR/MWh premium is charging for risk it can hedge for free; the defensible premium is about 10.5.
3. **Wind and solar have different risks.** Solar: profile. Wind: volume. They belong in different structures (solar: PaP or baseload with a shape adjustment; wind: volume protections or a portfolio).
4. **The clause is not free.** 17.5% of Borrentin's output falls at negative prices in 2027, about 30% over ten years; a buyer asking for the clause must pay for it in the strike.
5. **The biggest uncertainty is the future of cannibalisation,** not the weather: the solar and storage build-out after 2027 moves the ten-year price from 17.9 to 30.1 EUR/MWh, about as much as the risk premium and the costs together (13.7).

6.2 What this means for the rest of the project

- **Step 8** (hedging): the producer's remaining merchant exposure (if it keeps part of the output) and the buyer's baseload hedge, with market forwards instead of one assumed number.
- **Step 9** (public MVP): `ppa_pricing.json` feeds the dashboard and the one-page memo.

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7. Exercises

Warm-up (in `exercises/step07_pricing.py`, checked by `.venv/bin/pytest exercises -q`)

1. **How much risk does a PPA take away?** Merchant and pay-as-produced CFaR95 from the Step 6 scenario file.
2. **One price for ten years,** with a capture-rate trend, degradation and discounting.
3. **The buyer's premium,** with and without a baseload hedge, on a toy market.
4. **Written:** the price negotiation between 21.4 and 35.1 EUR/MWh.

Core

5. **The fair strike with the clause on scenarios.** Solve "PPA revenue with the clause = merchant revenue" over the scenarios (your Step 3 Exercise 3, now in expectation). How far above the plain 2027 fair price of 77.5 is it?
6. **A premium at a different quantile.** Recompute the premia at 10% and 1%. Which park's premium grows faster, and why?
7. **A zero-cost collar.** With a floor at 0 EUR/MWh for solar, find the cap C at which the cap is worth as much as the floor (METHODOLOGY section 4), so the producer pays nothing net.

Stretch

8. **Multi-year premium.** Simulate ten consecutive years by drawing ten weather years and price paths per scenario. Is the ten-year premium per MWh smaller or larger than the one-year one?

9. **Capital.** If the buyer holds capital equal to its premium at 10% per year for ten years, what does that add to the price?

▼ **Hints and solution sketches**

1. Linked rows, `K = rev.sum() / mwh.sum()`, PPA revenue `K * mwh`, `mean − quantile(0.05)`: EUR 3.19 m vs 0.70 m for Borrentin (the file's merchant keeps producing at negative prices and uses analog days from the whole window).
2. Loop over the years; Borrentin's 2027 numbers (fair 77.53, forward 141.06) with a −1 pp drift and 0.5% degradation at 6%: 71.91 EUR/MWh.
3. `buyer_value` by hand; the toy gives 28.61 without and 9.64 with the hedge.
4. The 13.7 EUR/MWh are the premium after the hedge (10.5) and the costs (3.2). Concede the hedged premium, not the unhedged one: the buyer can sell a baseload forward. Then argue the solar and storage build-out: storage alone moves the ten-year price by up to 8.7 EUR/MWh.
5. `K_clause = E[Σ q P 1(P ≥ 0)] / E[Σ q 1(P ≥ 0)]` (what a curtailing merchant earns per MWh it delivers); for Borrentin 97.76 against 77.53 EUR/MWh in 2027.
6. `risk_premia(P, Q, quantile=0.10)`.
7. Cap value E[Σ q (P − C)⁺] / E[Σ q] falls as C rises; bisect C until it equals the floor value (Borrentin 3.10 EUR per MWh of output, the value of curtailing).
8. Use `build_scenarios` with different seeds and sum over years before taking the quantile.
9. 10% × premium-at-risk per MWh, discounted.

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8. Self-check quiz

1. Under a pay-as-produced PPA, which risks does the producer keep and which does the buyer take?
2. Define CFaR95. Why do lenders and CFOs like it?
3. What is the buyer's value of a PaP PPA per scenario, and why is it zero on average at the fair price?
4. Why must the premium be computed after a baseload hedge?
5. How is the premium split into profile and volume, and why is solar's mostly profile?
6. Write the price stack from the forward to the indicative price.
7. How does a ten-year price follow from the first year's price, and when are they equal?
8. Why does a negative-price clause cost the solar park so much?
9. Which three assumptions move the solar price most, and by how much?
10. Why is the analog pool a configuration value?

▼ **Answers**

1. Producer: volume. Buyer: price level, shape (profile) and volume.
2. Expected revenue minus its 5% quantile: the cash shortfall of a one-in-twenty year, which is what debt service and budgets must survive.

3. Σ q (P − K): it pays K for each MWh and sells at P. At K* = E[Σ q P]/E[Σ q] the expectation is zero.
4. The price level is hedgeable with a baseload forward; charging for it would price a risk the buyer can remove almost for free.
5. Profile: expected hourly volume in every scenario; volume = joint − profile. Borrentin's volume barely varies between years (CoV 4.6%), while the depth of the midday dip varies a lot.
6. Forward × capture rate = fair price; minus risk premium, balancing and route to market, credit and liquidity = indicative price.
7. Energy- and discount-weighted average of yearly fair prices; equal when forward and capture rate are flat.
8. 17.5% of Borrentin's output falls in negative-price hours in 2027 (about 30% over ten years), which the clause leaves unpaid.
9. The solar and storage build-out after 2027 (–3.4 to +8.7 EUR/MWh across the scenarios), the forward (–5.6 / +5.5 for ±10%) and a capture-rate trend on top (±3.5 per pp a year).
10. It is a judgement that moved the solar price by several EUR/MWh until v1.7; it must be visible and challengeable, not buried in code. Since v1.7 it is also tested: moved to 2027 by the build-out trend, both pools give the same price.

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9. Interview corner

Your 60-second story (STAR):

> *Situation:* origination needed a defensible ten-year price for a PPA from a 114 MWp solar park and a statement of the risk we keep. *Task:* price the standard structures on joint weather and price scenarios and build the price from the forward to an offer. *Action:* I reused the tested contract functions on 496 scenarios, measured the producer's cash flow at risk per structure, computed the buyer's premium after a baseload hedge and split it into profile and volume, and added a ten-year layer with solar and storage build-out scenarios. *Result:* a fair price of 35.1 and an indicative 21.4 EUR/MWh; the PPA cuts our CFaR by 77%; and I showed that the storage build-out moves the price (17.9 to 30.1) about as much as the whole risk premium and costs.

Likely questions and strong answers:

- "*How do you price a pay-as-produced PPA?*" Forward times expected capture rate from joint weather–price scenarios, minus a premium for the unhedgeable profile and volume risk, minus balancing, route-to-market and credit costs; then a term structure for multi-year deals.
- "*What risk premium would you accept?*" The premium after the buyer hedges the level with baseload forwards; here about 10.5 EUR/MWh for solar, 5.5 for wind.
- "*What does the PPA do to our risk?*" It replaces price and shape risk with a fixed price; we keep volume risk (CFaR from EUR 3.02 m to 0.70 m per year for Borrentin).
- "*What would make you change the price?*" A view on future solar capacity and batteries (the build-out scenarios), and the forward curve on the day of signing (Step 8).

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10. How this maps to real work at Encavis

In this project	In a company
Fair price from 496 joint scenarios	PPA pricing models with weather years and price scenarios, often from vendors
Premium after a baseload hedge, 5% quantile	Risk premia from desk hedging costs, internal capital charges, market quotes
Balancing, credit as stated costs	Route-to-market contracts, balancing cost models, counterparty credit assessment
Ten-year price with solar and storage build-out scenarios	Long-term price curves and cannibalisation scenarios per technology
CFaR per structure	Budget at risk, debt sizing, the risk committee's view of a deal
`ppa_pricing.json`	Deal sheets and pricing memos for the investment committee

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11. Your to-dos for Step 7

- □ Read sections 1 and 4. By hand: forward 141.1, capture rate 0.55, premium 10.5, costs 3.2 → indicative price for 2027? (63.8 EUR/MWh.)
- □ Run section 5; in the notebook, find the strike in 7.3 at which the buyer's 5% quantile is zero.
- □ Exercises 1-4 in `exercises/step07_pricing.py`; check them in "My solutions: Step 7".
- □ **Commit Step 7** (commands below).
- □ **German vocabulary of this step:** der Strombezugsvertrag (PPA), der Festpreis, der Abnehmer, der Erzeuger, die Risikoprämie, das Profilrisiko, das Mengenrisiko, die Absicherung (hedge), der Terminkontrakt, Cashflow at Risk, der Kreditaufschlag, die Laufzeit, die Abzinsung, die Kannibalisierung.
- □ Stand-up update: "*Yesterday: priced the PPAs on the joint scenarios: Borrentin 35.1 fair, 21.4 indicative for ten years; the PPA cuts our cash flow at risk by 77%. Today: the public MVP (Step 9). Blockers: none.*"

Commit (check `git status` first: no `.env`, no `M .env.example`):

```
cd ".../Encavis quantitative/ppa-lab"
git status --short
git add src tests exercises config reports docs notebooks METHODOLOGY.md README.md
CHANGELOG.md Makefile pyproject.toml
git status --short
git commit -m "feat(pricing): PPA prices, risk premia and cash flow at risk on joint scenarios
(PPA-7)"
git push
gh run watch
```

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12. Glossary

Term	Meaning
Cash flow at risk (CFaR95)	Expected cash flow minus its 5% quantile
Risk premium	Discount on the fair price a risk taker asks for; here the 5% shortfall per MWh
Baseload hedge	Selling a constant band of power forward to fix the price level
Profile risk	Risk from the shape of output against the shape of prices
Volume risk	Risk from how much is produced, including its link to prices
Price stack	The layers from a forward price to an indicative PPA price
Tenor	Duration of a contract
Degradation	Annual loss of output of PV modules
Capture-rate trend	Expected change of the capture rate per year (cannibalisation vs storage)
Negative-price clause	No payment for output at negative prices; the producer curtails
Floor	Option that guarantees a minimum price

13. Sources

- Jorion, P. (2007): *Value at Risk*, 3rd ed. McGraw-Hill (VaR and cash-flow-at-risk concepts).
- RiskMetrics Group (1999): *CorporateMetrics Technical Document* (cash flow at risk).
- Jordan, D. C. and Kurtz, S. R. (2013): Photovoltaic degradation rates: an analytical review. *Progress in Photovoltaics* 21(1), 12-29 (median about 0.5% per year).
- Hirth, L. (2013): The market value of variable renewables. *Energy Economics* 38, 218-236.
- Industry PPA market reports (for example Pexapark, LevelTen) for current European PPA prices and structures.
- Encavis preparation guide: price stack (illustrative costs), PPA pricing interview questions.